Progress Report: Topological Cargo Stratification and Orthogonal Validation (Next Steps #4)

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1. Overview

Empirical Bayes linear modeling (limma-trend, $\mathrm{FDR} < 0.10$, $|\log_2 FC| \geq 1.0$) identified 42 differentially internalized extracellular vesicle (EV) cargo proteins across the quantified proteome, comprising 19 targets enriched under parental PC3 uptake and 23 targets enriched under metastatic PC3M uptake. To determine the structural localization of these candidates within the vesicle architecture and verify that they represent authentic EV constituents rather than co-isolated cellular debris, a two-stage characterization pipeline was executed. First, quantified proteins were stratified into discrete topological compartments using reference surfaceome annotations. Second, all significant candidates were cross-validated against published human EV proteomes in the *Vesiclepedia* compendium.

2. Topological Cargo Stratification

To distinguish surface-accessible interaction drivers from encapsulated internal cargo, topological annotations were derived from the *In Silico Human Surfaceome* repository (SURFY; Bausch-Fluck et al., *PNAS* 2018). Predicted transmembrane (TM) alpha-helix counts and controlled UniProt ontological keywords were extracted from the "SurfaceomeMasterTable" sheet-reference. For multigene families (Family_*), surface classification required unanimous consensus across all constituent paralogs.

Each protein was evaluated sequentially through a priority-ordered pipeline:

1. **"Membrane":** Assigned to proteins achieving unanimous consensus in the SURFY *in silico* surfaceome reference.
2. **"Extracellular":** Assigned to non-transmembrane proteins ($\mathrm{TM} = 0$) harboring at least one extracellular matrix, basement membrane, or secreted keyword.
3. **"Organellar":** Assigned to proteins excluded from SURFY that either carry "cell membrane" or "membrane" annotations paired with lipid-anchor modifications, or contain predicted transmembrane domains ($\mathrm{TM} > 0$) without nuclear localization tags.
4. **"Intracellular":** Assigned to transmembrane proteins ($\mathrm{TM} > 0$) carrying nuclear or chromatin annotations, as well as non-transmembrane proteins ($\mathrm{TM} = 0$) matching intracellular cytoplasmic, cytoskeletal, or organellar lumen keywords.
5. **"Ambiguous" / "Undefined":** Assigned as a fallback state when multigene family members exhibit partial SURFY consensus ("Ambiguous") or when no surfaceome, transmembrane, or ontological keywords are matched ("Undefined").

Table 1. Deterministic topological gating hierarchy and controlled UniProt keyword dictionaries in priority order.

Class	Classification Rule	Controlled Keywords (Tags)
1. Membrane	Unanimous SURFY surfaceome match.	*Not applicable*
2. Extracellular	Non-SURFY, $\mathrm{TM} = 0$, and ≥ 1 extracellular tag.	extracellular matrix, secreted, basement membrane, cytokine, growth factor
3. Organellar	Non-SURFY with ≥ 1 membrane/lipid tags, or $\mathrm{TM} > 0$ (non-nuclear).	cell membrane, membrane + (lipoprotein, prenylation, palmitate, myristate, gpi-anchor)
4. Intracellular	$\mathrm{TM} > 0$ with ≥ 1 nuclear tags, or $\mathrm{TM} = 0$ with ≥ 1 intracellular tag.	cytoplasm, cytoskeleton, nucleus, chromosome, nucleosome core, endoplasmic reticulum, golgi apparatus, mitochondrion, mitochondrion inner membrane, microsome, ribosome, ribosomal protein, ribonucleoprotein, chaperone, annexin, calcium/phospholipid-binding, glycolysis, pentose shunt, gluconeogenesis, spliceosome, mrna processing, mrna splicing, dna repair, dna replication
5. Ambiguous / Undefined	Partial SURFY match (Ambiguous) or no rules met (Undefined).	*Not applicable*

Automated hierarchical gating classified 38 of the 42 statistically significant targets. Four remaining candidates (Family_ATP_3, H3-7, CPNE3, and TGM2) initially fell into the Undefined category due to legacy 2018 HGNC nomenclature in the SURFY reference (ATP5MC1–3 indexed as ATP5G1–3; H3-7 as HIST2H3PS2) or the absence of subcellular localization keywords in the 2018 UniProt snapshot (CPNE3, TGM2). Based on current canonical annotations in UniProtKB, three candidates were assigned to the **Intracellular** compartment (mitochondrial ATP synthase subunits, a core nucleosomal histone, and a cytosolic calcium-dependent phospholipid-binding protein, respectively), whereas TGM2 (a non-classically secreted extracellular matrix cross-linker) was assigned to the **Extracellular** class.

Across the 42 resolved drivers, intracellular, membrane, and organellar categories were balanced between phenotypes:

- **Intracellular:** 7 in PC3 vs. 7 in PC3M (14 total)
- **Membrane:** 6 in PC3 vs. 5 in PC3M (11 total)
- **Organellar:** 5 in PC3 vs. 5 in PC3M (10 total)
- **Extracellular:** 1 in PC3 vs. 6 in PC3M (7 total)

In contrast to the symmetric distribution of internal and membrane-spanning proteins, metastatic PC3M vesicles exhibited a six-fold enrichment in Extracellular matrix and corona components, representing the main structural divergence between the two cargo profiles.

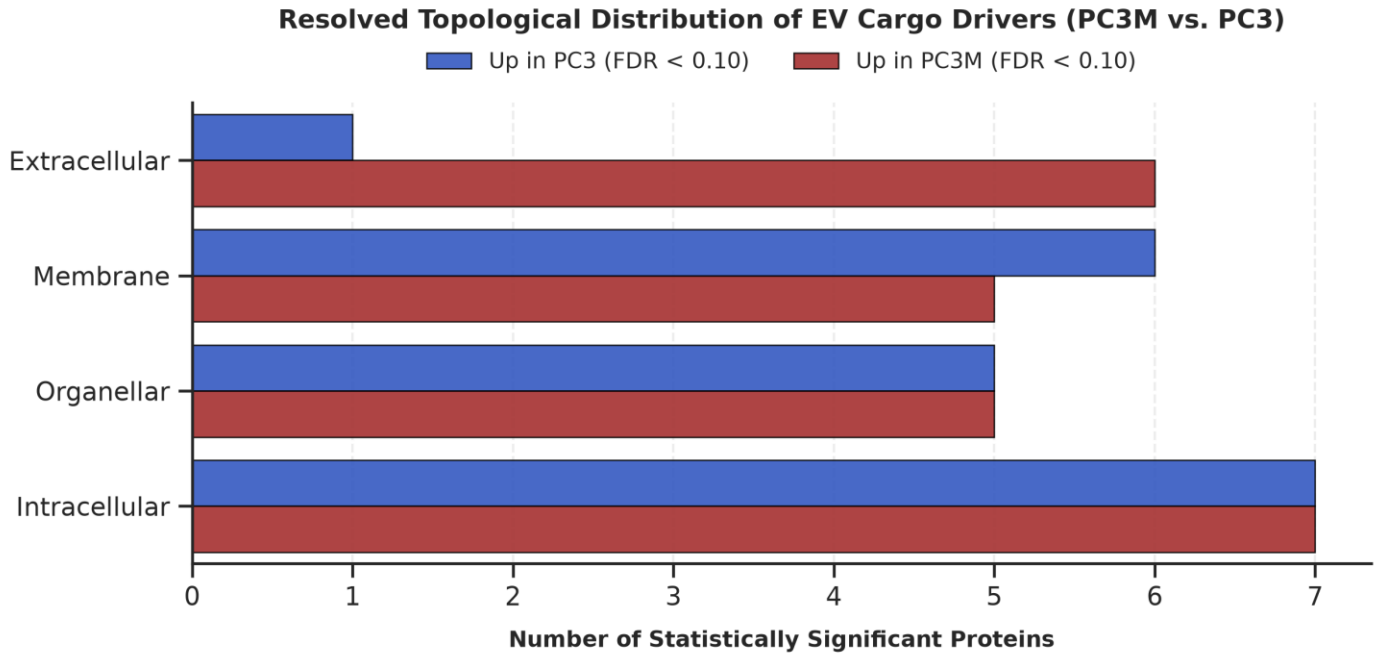


Figure 1. Resolved topological distribution of EV cargo drivers (PC3M vs. PC3). Horizontal grouped bar chart illustrating the distribution of the 42 statistically significant cargo proteins (FDR < 0.10) across four structural compartments following targeted curation of legacy-symbol entries. Blue bars denote proteins enriched under parental PC3 uptake ($n = 19$), and red bars denote proteins enriched under metastatic PC3M uptake ($n = 23$). While intracellular, organellar, and integral membrane proteins are comparably represented across both conditions, metastatic PC3M vesicles show a selective enrichment of extracellular matrix and EV corona proteins (6 in PC3M vs. 1 in PC3).

3. Orthogonal Validation

To verify whether the 42 differential drivers represent established vesicular proteins rather than experimental artifacts, candidates were queried against the *Vesiclepedia* repository (Version 5.1, 2023 release). Reference records were restricted to human protein identifications (SPECIES == “Homo sapiens”, CONTENT TYPE == “protein”), and support was quantified as the number of unique published studies (EXPERIMENT ID) reporting each target.

To maintain consistency with the peptide grouping strategy, multigene families (Family_*) were evaluated using a set-union rule: all unique EXPERIMENT ID records across constituent paralogs were combined into a single non-redundant set. This approach captures every independent study reporting at least one member of the protein family while preventing duplicate counting within the same publication.

Initial matching using current HGNC symbols confirmed 36 of the 42 candidates (86%). The remaining six candidates (H4C1, CCN1, H3-3A, MACROH2A2, Family_ATP_3, and H3-7) returned zero matches solely due to 2018 – 2020 HGNC nomenclature revisions relative to historical proteomics datasets. Querying their legacy official symbols (HIST1H4A–F, CYR61, H3F3A, H2AFY2, ATP5G1–3, and HIST2H3PS2) via the same rule recovered literature support for all six targets. After accounting for legacy HGNC aliases, all of the statistically significant cargo drivers were independently confirmed as documented human EV constituents.

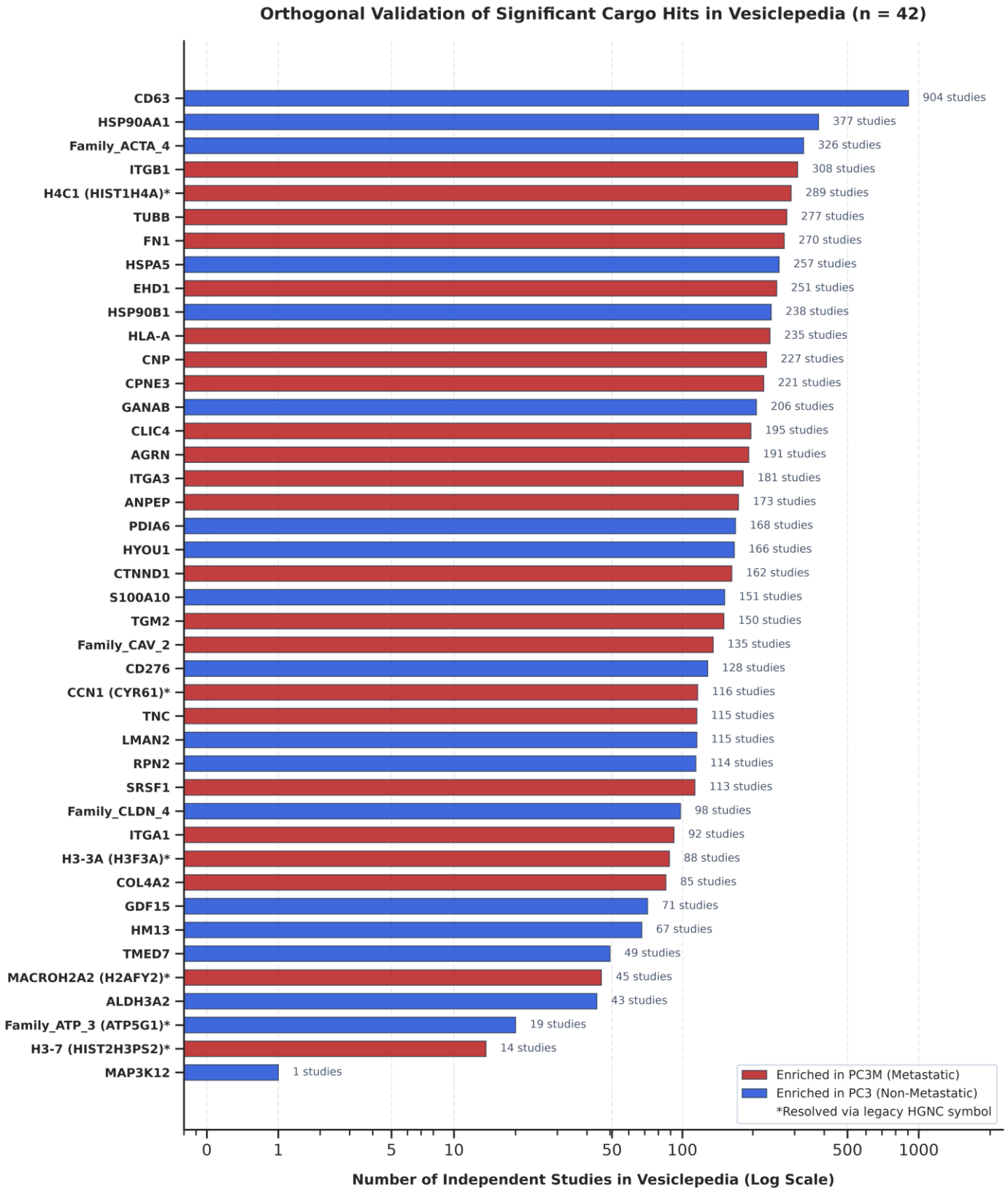


Figure 2. Orthogonal validation of significant cargo hits in Vesiclepedia ($n = 42$). Horizontal bar chart displaying the number of independent published human EV studies supporting each of the 42 statistically significant cargo drivers in Vesiclepedia (Version 5.1; logarithmic X-axis). Bars are colored by uptake phenotype (red: enriched in metastatic PC3M, $n = 23$; blue: enriched in parental PC3, $n = 19$). Asterisks (*) indicate six candidates resolved via legacy HGNC gene symbols (HIST1H4A–F, CYR61, H3F3A, H2AFY2, ATP5G1–3, and HIST2H3PS2). Across all 42 targets, 100% are corroborated by independent human EV proteomics studies, confirming the biological authenticity of the isolated cargo panel.

Up in PC3 (FDR < 0.10)									Up in PC3M (FDR < 0.10)								
Gene / Gene set	limma		ranks			surfaceome			Gene / Gene set	limma		ranks			surfaceome		
	Log2FC	q_value	Delta_PC3	Delta_PC3M	Delta_Delta_Rank	Is_Surfaceome	TM_Domains	Verdict		Log2FC	q_value	Delta_PC3	Delta_PC3M	Delta_Delta_Rank	Is_Surfaceome	TM_Domains	Verdict
MAP3K12	-2.25	0.003	+127	+44	-83	FALSE	0	Intracellular	COL4A2	+6.01	0.002	-110	+94	+205	FALSE	0	Extracellular
HM13	-1.77	0.001	+100	+0	-100	TRUE	9	Membrane	TGM2	+4.12	0.001	-43	+48	+92	FALSE	0	Extracellular
LMAN2	-1.71	0.009	+74	+22	-52	TRUE	1	Membrane	FN1	+3.76	0.008	0	+67	+67	FALSE	0	Extracellular
GDF15	-1.47	0.012	+44	-5	-49	FALSE	0	Extracellular	TNC	+2.10	0.020	+23	+78	+55	FALSE	0	Extracellular
S100A10	-1.47	0.019	+52	-29	-81	FALSE	1	Organellar	ITGA1	+2.01	0.008	+40	+100	+60	TRUE	1	Membrane
Family_ATP_3	-1.39	0.051	+46	+14	-32	FALSE	0	Intracellular	CTNND1	+1.97	0.001	-59	+37	+96	FALSE	0	Organellar
HSP90B1	-1.35	0.004	+48	+12	-36	FALSE	0	Intracellular	EHD1	+1.81	0.093	+32	+112	+80	FALSE	0	Organellar
CD276	-1.26	0.042	+13	-26	-39	TRUE	1	Membrane	Family_CAV_2	+1.79	0.002	-11	+31	+42	FALSE	0	Organellar
CD63	-1.25	0.009	+62	-9	-70	TRUE	4	Membrane	ANPEP	+1.79	0.005	+1	+109	+108	TRUE	1	Membrane
ALDH3A2	-1.23	0.017	+20	-30	-50	FALSE	1	Organellar	CCN1	+1.70	0.012	+1	+105	+104	FALSE	0	Extracellular
HYOU1	-1.22	0.044	+27	-18	-45	FALSE	0	Intracellular	ITGA3	+1.54	0.009	+14	+113	+100	TRUE	1	Membrane
RPN2	-1.21	0.014	+101	+27	-75	FALSE	3	Organellar	MACROH2A2	+1.48	0.026	-64	+3	+67	FALSE	1	Intracellular
HSP90AA1	-1.15	0.017	-6	-57	-51	FALSE	0	Organellar	TUBB	+1.39	0.027	-117	-45	+72	FALSE	0	Intracellular
Family_CLDN_4	-1.13	0.036	+2	-51	-52	TRUE	4	Membrane	ITGB1	+1.38	0.025	-3	+34	+37	TRUE	1	Membrane
GANAB	-1.12	0.028	+48	-14	-62	FALSE	0	Intracellular	H3-3A	+1.37	0.033	-52	+9	+61	FALSE	0	Intracellular
Family_ACTA_4	-1.10	0.030	+32	-11	-43	FALSE	0	Intracellular	CLIC4	+1.35	0.025	-11	+12	+23	FALSE	0	Organellar
HSPA5	-1.05	0.005	+6	+0	-5	FALSE	0	Intracellular	H3-7	+1.27	0.045	-72	-3	+69	FALSE	0	Intracellular
PDIA6	-1.03	0.026	+90	+47	-43	FALSE	0	Organellar	CPNE3	+1.19	0.018	-54	+10	+64	FALSE	0	Intracellular
TMED7	-1.01	0.069	+76	+10	-66	TRUE	1	Membrane	H4C1	+1.15	0.017	-7	+3	+10	FALSE	0	Intracellular
									AGRN	+1.06	0.012	-14	+33	+47	FALSE	0	Extracellular
									SRSF1	+1.04	0.009	-56	+18	+74	FALSE	0	Intracellular
									HLA-A	+1.03	0.017	+15	+70	+55	TRUE	1	Membrane
									CNP	+1.02	0.030	-32	+17	+49	FALSE	0	Organellar

Figure 3. Integrated multi-metric summary of differentially internalized EV cargo drivers ($n = 42$). Side-by-side overview of all statistically significant cargo targets ($\mathrm{FDR} < 0.10$, $|\log_2 \mathrm{FC}| \geq 1.0$) identified by empirical Bayes linear modeling, partitioned into parental PC3-enriched ($n = 19$, left) and metastatic PC3M-enriched ($n = 23$, right) modules and ranked by effect size magnitude ($\log_2 \mathrm{FC}$). Gene identifiers encompass both unique HGNC symbols and resolved multigene families (Family_*). For each candidate, three analytical layers are summarized: parametric statistics (limma), reporting relative uptake fold change ($\log_2 \mathrm{FC}$, shaded from blue to red) and Benjamini-Hochberg false discovery rate (q_value); non-parametric rank shifts (ranks), displaying condition-specific hierarchy shifts ($\Delta\mathrm{Rank}_{\mathrm{PC3}}$, $\Delta\mathrm{Rank}_{\mathrm{PC3M}}$) alongside the cross-phenotype contrast ($\Delta\Delta\mathrm{Rank}$) to illustrate directional concordance with parametric effect sizes; and topological classification (surfaceome), detailing consensus presence in the *in silico* human surfaceome (Is_Surfaceome), predicted transmembrane helix count (TM_Domains), and the final curated structural compartment (Verdict: *Extracellular*, *Membrane*, *Organellar*, or *Intracellular*).